

JOANNA SIERRA-MENDOZA

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Education

Western Governors University

January 2025 - Expected May 2028

Bachelor of Science in Software Engineering

Bellevue University

August 2023 - May 2026

Master of Science in Data Science

Concordia University

August 2016 - May 2020

Bachelor of Science in Biology

Technical Skills

Programming Languages: Python, R, SQL, JavaScript

Web Technologies: HTML, CSS

Frameworks & Libraries: Angular, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Tools & Technologies: Git, GitHub, Docker, AWS, Jupyter Notebook, Power BI

Big Data Technologies: HDFS, HBase, Spark, Kafka, Solr, NiFi

Experience

Handshake AI

November 2025 - Present

AI Trainer

- Contribute to multiple AI training projects by reviewing model outputs for reasoning quality, accuracy, safety, and alignment with project guidelines.
- Provide structured feedback and quality evaluations to improve AI model performance, reliability, and overall user experience.

Outlier AI

June 2024 - November 2025

AI Trainer

- Analyzed model outputs using structured evaluation rubrics, contributing to dataset quality improvements.
- Collaborated on diverse AI training tasks involving prompt evaluation, response assessment, and quality review across different projects.

Zoetis Inc

December 2022 - September 2024

Laboratory Technician (mAbs)

Lincoln, NE

- Monitored and analyzed cell growth data while creating trend reports to support process optimization.
- Served as Environmental Monitoring Reporter, analyzing and reporting environmental data to maintain quality and compliance standards.
- Contributed to the revision and improvement of batch records, SOPs, and OJTs to enhance accuracy and operational efficiency.

Projects

Credit Risk MLOps Pipeline | Python, XGBoost, FastAPI, AWS, Docker, MongoDB

April 2026

- Built an end-to-end MLOps pipeline to predict credit card default risk from financial transaction data.
- Implemented data ingestion from MongoDB and designed a modular pipeline for validation, transformation, and model training.
- Developed a FastAPI web application with an interactive HTML form to collect user inputs and generate real-time predictions.
- Containerized the application with Docker and deployed to AWS (ECR, EC2) with CI/CD automation using GitHub Actions.

Alabama City Guide Website | HTML, CSS, JavaScript

August 2025

- Built a responsive multi-page website with structured navigation and organized content across multiple city pages.
- Designed and styled layouts using CSS to ensure visual consistency and responsiveness across different screen sizes.

Credit Card Fraud Detection Model | Python, Pandas, Scikit-learn, Matplotlib

May 2025

- Performed exploratory data analysis on a large-scale credit card transactions dataset to identify patterns, class imbalance, and feature distributions.
- Implemented data preprocessing and train-test-validation splits to ensure reliable model evaluation and minimize overfitting.
- Improved fraud detection performance by balancing the dataset and evaluating multiple machine learning models.